

# Marcus Ruth

[polter.sh](#) | [linkedin.com/in/marcusruth](#) | [github.com/polter-dev](#)

## EDUCATION

### University of Central Florida

*Bachelor of Science in Computer Science – Accelerated BS to MS Program*

Aug. 2024 – May 2027

Orlando, FL

### Valencia College

*Associate of Arts – General Studies*

Aug. 2022 – May 2024

Orlando, FL

## RELEVANT EXPERIENCE

### University of Central Florida

*Teaching Assistant - Data Structures & Algorithms (COP 3502C) and Machine Learning (CAP 4611)*

Jan. 2025 – Present

Orlando, FL

- Lead **lab sessions and office hours** covering data structures (trees, heaps, hash tables) and ML topics including classification pipelines, ensemble methods, clustering, and **neural networks (NN, CNN, DNN)**.
- **Designed C assignments** on **dynamic memory allocation** and **stack vs. heap management**, including a BST with augmented subtree sizes and a recursive permutation problem.
- Grade assignments and provide constructive feedback to help students improve algorithmic problem-solving and debugging of **C-based implementations**.

### Mindhome

*Software Engineer - Internship*

Dec. 2022 – Feb. 2023

Remote

- Completed **ROS2** training utilizing **Python**, building foundational skills in AI and robotics programming.
- Contributed to backend development in **Ruby**, implementing **API endpoints** and assisting with **database integration** to support core platform functionality.
- Collaborated with senior engineers in **code reviews** and **Agile** workflows to improve system reliability.

## PROJECTS

### Sedge | *Swift, SwiftUI, Plaid API, Python, LLMs/SLMs, Jev* | *Graduate AI Course Project*

Sep. 2026

- Building an iOS **banking AI agent** that uses the free **Plaid API** to answer questions about spending.
- Designing a **hybrid SLM/LLM** setup: an **on-device small language model** answers first and escalates to a **cloud LLM** when unsure, benchmarked on **670+ synthetic transactions**.
- Integrating TypeSafe's **Jev** classifier for tool routing and escalation; early tests routed **11/11** answerable queries to the correct tool in **~0.25s** while sending only question text, never financial data.

### Encore | *React Native, Whisper, MobileCLIP, Qwen3, RAG, LoRA* | *UCF Capstone*

Aug. 2026

- **Project manager and lead AI engineer** for an app that recalls your day, like “where are my keys?”
- Designed the AI pipeline: **Whisper** transcription, **MobileCLIP2** image embeddings, a **Qwen3-0.6B** moment classifier, and **SLMs** querying a self-evolving **EvoGraph-R1** knowledge graph.
- Architected a **model cascade** where a lightweight trigger model runs continuously and calls heavier embedding and generation models only when needed, keeping inference cheap enough for on-device use.
- Benchmarking **RAG** retrievers (**BGE, E5, GTE, GraphRAG**) on accuracy, latency, and memory, and **fine-tuning** small models with **LoRA** and distillation to improve on-device performance.

### Terminal Presence | *Go, Discord IPC, GoReleaser, systems programming* | [GitHub](#)

July 2026

- Shipped an open-source **cross-platform Go CLI** that shows your active terminal program as a live **Discord Rich Presence** status; **41 user downloads** and **7 GitHub stars** across **5 releases**.
- Detects **48 built-in tools** by scanning running processes, using local IPC with **no bot or token**.
- Supports **hot-reloaded TOML config** and custom tools; distributed via **Homebrew, Scoop, and .deb/.rpm packages** with start-at-login on macOS, Linux, and Windows.
- Wrote **22K+ lines of Go** with **960+ tests**, a checksum-verified installer, and a guided setup wizard.

## TECHNICAL SKILLS

**Languages:** Python, Java, C/C++, Go, Swift, TypeScript, JavaScript, SQL

**Web & Mobile:** Next.js, React, React Native, SwiftUI, tRPC, Zod, Supabase, Postgres

**Libraries:** PyTorch, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Nixtla, Hugging Face, Transformers

**AI/ML:** Classification, Regression, Ensembles, Clustering, CNNs/DNNs, Hyperparameter Tuning, NLP, RAG, LLMs

**Developer Tools:** Claude Code, Codex, Git, VS Code, Unix/Linux, AWS, PostgreSQL, Neovim